

Mogalapalli Sohan

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Profile

Computer Science and Engineering student at Vellore Institute of Technology passionate about Artificial Intelligence, Distributed Systems, Cloud Computing, and Software Engineering. Experienced in building AI-powered applications, agentic workflows, retrieval-augmented generation systems, and decentralized cloud storage architectures while continuously exploring scalable software solutions.

Education

Vellore Institute of Technology, Vellore

2024 – 2028

Bachelor of Technology in Computer Science and Engineering (Core)

CGPA: 9.49 / 10.0

Relevant Coursework

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Artificial Intelligence, Cloud Computing

Projects

CloudVaultX – AI-powered Decentralized Cloud Storage Architecture

Ongoing

- Developing a decentralized cloud storage platform that securely distributes encrypted file chunks across multiple storage nodes.
- Implementing AI-based node selection to optimize storage allocation, availability, and retrieval performance.
- Designing metadata management, secure chunking, encryption, and fault-tolerant storage mechanisms.
- Designing scalable distributed storage architecture with secure metadata management, replication, and fault-tolerant data retrieval.
- Tech Stack:** Python, FastAPI, LangGraph, AWS, SQLite, AES Encryption

AI Software Engineering Copilot for Large Codebases

Ongoing

- Building an AI-powered software engineering assistant for understanding and navigating large code repositories.
- Implementing Retrieval-Augmented Generation (RAG), semantic search, and intelligent code summarization.
- Developing agentic workflows using LangChain and LangGraph to automate software engineering tasks.
- Improving developer productivity through repository-wide question answering and contextual code assistance.
- Tech Stack:** Python, LangChain, LangGraph, Ollama, FAISS, RAG

Real-time Audio-In, Audio-Out Conversational AI Assistant

- Developed a real-time conversational AI assistant supporting speech-to-text and text-to-speech interactions.
- Integrated Large Language Models (LLMs) to generate context-aware conversational responses.
- Implemented low-latency streaming audio pipeline for seamless voice interactions.
- Tech Stack:** Python, Ollama, Faster-Whisper, sounddevice

Smart Home Security System Using Ultrasonic Sensor

Mini Project

- Designed and implemented an Arduino Uno-based smart home security prototype using an HC-SR04 ultrasonic sensor.
- Developed real-time intrusion detection with automatic alarm activation using buzzer and LED indicators.
- Tech Stack:** Arduino Uno, C++, HC-SR04, Arduino IDE

Technical Skills

Languages	C, C++, Java, Python, JavaScript
Frontend	React.js, HTML, CSS
Backend	Node.js
Databases	MySQL, MongoDB, SQL*Plus
Cloud	AWS
AI & Machine Learning	LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Ollama, Vector Databases
Tools & Platforms	Git, GitHub, VS Code, Arduino IDE

Certifications

- IBM Developer Skills Network – **Building Agents with Agentic AI** (July 2026)

Achievements

- Qualified for **Round 2 of Flipkart GRiD**, demonstrating strong problem-solving and software engineering aptitude.